

TRON-ICE: Machine Learning-Enhanced Transaction Matching for Tracing Illicit Fund Flows Through Instant Cryptocurrency Exchanges on the Tron Network

Group 13 University of Information Technology (UIT), Vietnam National University Ho Chi Minh City, Vietnam

Abstract—Instant Cryptocurrency Exchange (ICE) services have emerged as a primary vector for cross-chain asset obfuscation and illicit fund laundering, driven by the absence of stringent Know Your Customer (KYC) protocols and obscured on-chain transaction metadata. While existing literature has predominantly examined ICE ecosystems within Ethereum and EVM-compatible environments, the Tron network—despite processing the lion’s share of global USDT stablecoin volume and being heavily exploited for illicit financial flows—remains critically under-analyzed. In this paper, we propose **TRON-ICE**, a novel blockchain forensics framework that extends ICE transaction tracking to the Tron blockchain. Moving beyond rigid, threshold-dependent heuristic matching, **TRON-ICE** introduces a supervised machine learning approach leveraging Gradient-Boosted Decision Trees (XGBoost). By parsing on-chain transaction data extracted via the TronGrid API, we construct a comprehensive, labeled ground-truth dataset of deposit-withdrawal pairs. Our matching pipeline integrates multi-dimensional features, including temporal proximity with exponential time-decay, value deviation metrics, token consistency, and critical security indicators such as address reuse patterns and temporal behavioral habits. Empirical evaluations demonstrate that **TRON-ICE** significantly outperforms baseline heuristics, achieving higher F1-scores and robustness against dynamic service fee fluctuations. Finally, we present real-world case studies illustrating the framework’s efficacy in unmasking complex money laundering pipelines on Tron.

Index Terms—Instant Cryptocurrency Exchanges, Tron blockchain, anti-money laundering, machine learning, transaction matching, cross-chain tracing, TRC-20

I. INTRODUCTION

The Tron network has emerged as a dominant platform for stablecoin transactions, processing over 50% of global USDT volume as of 2024 [2] due to its high throughput and low transaction fees. However, these identical attributes have attracted illicit actors exploiting the network for cost-efficient money laundering.

In particular, Instant Cryptocurrency Exchange (ICE) services—centralized, cross-chain platforms operating without strict KYC requirements—have become a preferred laundering channel following the 2022 OFAC sanctioning of Tornado Cash [3]. While Hu et al. [4] provided a foundational study on ICE workflows and proposed a heuristic algorithm to match deposits and withdrawals, their analysis explicitly excluded the Tron network. Furthermore, their approach relies on static thresholds for value deviation (ΔV) and time difference (ΔT),

which fail to adapt to Tron’s unique fee mechanisms (Energy/Bandwidth) and block dynamics. Recent studies [7], [8] suggest that machine learning can uncover complex patterns that these rigid, static rules miss.

To bridge these gaps, we present **TRON-ICE**, a framework tailored for detecting and analyzing ICE services on the Tron blockchain. This paper makes three primary contributions:

- 1) **Ecosystem Extension:** We conduct the first dedicated measurement of ICE services on Tron, leveraging a custom data collection pipeline via the TronGrid API to curate a novel ground-truth dataset of TRC-20 stablecoin laundering.
- 2) **Machine Learning-Driven Matching:** We replace rigid heuristic thresholds with an XGBoost classifier that dynamically learns optimal deposit-withdrawal matching boundaries by incorporating temporal features, value deviations, and a time-decay weighting scheme.
- 3) **Empirical Evaluation:** We evaluate our framework against historical ground-truth data, demonstrating substantial performance improvements over the baseline heuristic method, and validate its real-world utility through case studies of Tron-based money laundering incidents.

The remainder of this paper is organized as follows. Section II provides background on the Tron network and ICE services. Section III reviews related work. Section IV details the proposed TRON-ICE framework. Section V presents the experimental evaluation. Section VI describes case studies. Section VII discusses limitations and implications. Section VIII concludes the paper.

II. BACKGROUND

A. The Tron Blockchain

Tron is an account-based public blockchain that employs a Delegated Proof-of-Stake (DPoS) consensus mechanism with 27 Super Representatives producing blocks every 3 seconds [1]. Unlike Ethereum, which charges gas fees in Ether, Tron uses a dual resource model: Bandwidth (consumed by all transactions) and Energy (consumed by smart contract executions). Users can freeze TRX tokens to acquire these resources, effectively enabling zero-fee transactions for holders who maintain sufficient staked balances.

The TRC-20 token standard is analogous to ERC-20 on Ethereum and is the primary standard for stablecoin issuance on Tron. USDT on Tron (TRC-20 USDT) has become the largest stablecoin by transfer volume globally, surpassing its ERC-20 counterpart. This dominance is driven by significantly lower transfer costs (typically under \$1 compared to \$5–\$20 on Ethereum).

B. Instant Cryptocurrency Exchange Services

ICE services are centralized platforms that enable users to bridge cryptocurrency assets across blockchains without account registration or KYC verification [4]. The typical workflow involves three steps: (1) the user submits a request specifying source and destination tokens and provides a receiving address; (2) the service generates a deposit address to which the user sends funds; (3) upon confirmation of the deposit, the service transfers the equivalent value (minus a fee of 1–5%) to the user’s specified receiving address on the destination chain.

Hu et al. [4] classified ICE services into two types: Standalone services, which manage their own hot wallets for aggregating deposits and disbursing withdrawals, and Delegated services, which route funds through external services or centralized exchanges. The Standalone type is amenable to on-chain analysis because deposit aggregation and withdrawal disbursement patterns are directly observable.

C. Problem Statement

Given the set of deposit transactions $\mathcal{D} = \{d_1, d_2, \dots, d_n\}$ and withdrawal transactions $\mathcal{W} = \{w_1, w_2, \dots, w_m\}$ associated with an ICE service’s hot wallets on the Tron blockchain and supported chains, the matching problem is to find the correct pairing function $f : \mathcal{D} \rightarrow \mathcal{W} \cup \{\emptyset\}$ that maps each deposit to its corresponding withdrawal (or to $\emptyset$ if the withdrawal occurred on an unsupported chain). This paper addresses the problem of learning f from labeled examples using supervised machine learning, and of measuring the volume and patterns of illicit deposits in $\mathcal{D}$.

III. RELATED WORK

We organize related work into three thematic groups: ICE service analysis, blockchain-based anti-money laundering, and machine learning for cryptocurrency forensics.

A. ICE Service Analysis

The study of swap-based no-KYC services originated with Yousaf et al. [10] and Zhang et al. [11], who demonstrated that cross-service linking could de-anonymize users through heuristic matching algorithms on platforms like ShapeShift. Building upon this foundation, Hu et al. [4] provided the first systematic taxonomy of the broader ICE ecosystem and proposed a time-value heuristic matching baseline on Ethereum. However, these pioneering studies predominantly rely on rigid, rule-based heuristics ($V_{in} \approx V_{out}$) tailored for Ethereum-like fee structures. Consequently, they suffer from

TABLE I
COMPARISON OF RELATED APPROACHES.

Method	ICE	Tron	ML	Match	AML	Year
Yousaf et al. [10]	✓	×	×	×	×	2019
Zhang et al. [11]	✓	×	×	✓	×	2020
Hu et al. [4]	✓	×	×	✓	✓	2024
Tsai et al. [5]	×	✓	✓	×	✓	2024
Lam et al. [7]	×	✓	✓	×	✓	2024
Lin et al. [14]	×	✓	×	✓	✓	2025
TRON-ICE (Ours)	✓	✓	✓	✓	✓	2026

ICE = ICE service focus; Tron = Tron-specific analysis; ML = machine learning matching; Match = deposit-withdrawal matching; AML = anti-money laundering analysis.

significant performance degradation when applied to high-throughput, low-fee networks like Tron, where transaction values fluctuate dynamically due to unique Energy/Bandwidth fee mechanisms. Thus, adapting ICE analysis to Tron remains a critical, unsolved challenge.

B. Blockchain Anti-Money Laundering

Wu et al. [12] conducted a large-scale study of asset flows in Ethereum heists, tracking laundering patterns through mixing services, decentralized exchanges, and cross-chain bridges. Their work established a money flow tracking framework that combined address labeling with transaction graph analysis. Zhou et al. [13] proposed a visual analytics system for detecting money laundering patterns in cryptocurrency exchanges, employing network visualization to identify suspicious fund movements. Lin et al. [14] developed an automated cross-chain transaction uncovering system that traces fund flows across multiple blockchain ecosystems, addressing the challenge of fragmented evidence across chains. Tsai et al. [5] investigated illicit Tron wallets through temporal behavior analysis, finding that temporal activity patterns can distinguish malicious from benign wallets with high accuracy. Xia et al. [8] proposed a digital token transaction tracing method specifically for the Tron blockchain, addressing the challenge of following stablecoin flows through complex transfer chains.

C. Machine Learning for Cryptocurrency Forensics

While previous works by Pocher et al. [15] and Venckauskas et al. [16] successfully applied supervised learning (e.g., XGBoost) to detect illicit wallets, and Lam et al. [7] targeted USDT anomalies on Tron, they focus primarily on static wallet classification rather than dynamic transaction matching. None of these ML models have been adapted to solve the complex deposit-withdrawal linkage problem specific to ICE services.

As summarized in Table I, the current literature presents a noticeable gap: existing ICE matching techniques are confined to rigid heuristics on Ethereum [4], while advanced ML-based forensic tools on Tron [5], [7] strictly focus on wallet classification, ignoring the cross-chain money laundering facilitated by ICEs.

To bridge this gap, our paper introduces **TRON-ICE**. By replacing outdated deterministic rules with a supervised machine learning approach (XGBoost) tailored for the Tron network's unique fee dynamics, our work is the first to combine ICE service analysis, Tron blockchain forensics, and ML-enhanced deposit-withdrawal matching into a unified anti-money laundering framework.

IV. METHODOLOGY

A. Notation

Table II summarizes the notation used throughout this paper.

TABLE II
SUMMARY OF BASIC NOTATIONS AND SYSTEM ENTITIES

Symbol	Description
$\mathcal{D}$	Set of identified deposit operations
$\mathcal{W}$	Set of identified withdrawal operations
d_i	A deposit operation tuple: $(t_i^d, c_i^d, v_i^d, a_i^{\text{from}})$
w_j	A withdrawal operation tuple: $(t_j^w, c_j^w, v_j^w, a_j^{\text{to}})$
t_i^d, t_j^w	Block timestamps of the operations (in milliseconds)
c_i^d, c_j^w	Token types (e.g., TRX, USDT, USDC)
v_i^d, v_j^w	Dollar values of the operations computed at transaction time
$a_i^{\text{from}}, a_j^{\text{to}}$	Sender and recipient on-chain addresses, respectively
λ	Empirical time-decay parameter (set to 0.005)
y_{ij}	Ground truth binary label (1 for true match, 0 for negative sample)
$\hat{y}_{ij}$	Predicted matching probability generated by XGBoost

B. Data Collection

We collect on-chain data through the TronGrid API [19], which provides programmatic access to all transactions, TRC-20 token transfers, and internal transactions on the Tron network. For each target ICE service, we first identify the set of hot wallet addresses by analyzing known deposit addresses and tracing fund aggregation patterns, following the methodology established by Hu et al. [4]. We then collect all incoming and outgoing transactions of these hot wallets over the analysis period.

Each transaction is parsed into a standardized record containing: transaction ID, sender address, recipient address, token amount, token type, and block timestamp in milliseconds. Historical token prices are retrieved via the CoinGecko API [22] to compute the uniform dollar values (v) at transaction time.

To ensure high-fidelity value normalization across different time horizons, TRON-ICE integrates a dual-source price oracle framework. While CoinGecko API is utilized for macro-level historical pricing and cross-token consistency, Binance SPOT market data (specifically for the TRX/USDT pair) is ingested to provide high-granularity (minute-by-minute) price benchmarks. This dual-source approach mitigates the impact of intra-day volatility on the value deviation metric ΔV_{ij} .

TABLE III
ON-CHAIN TRANSACTION STATISTICS COLLECTED FROM THE TRON NETWORK (2024–2025)

Service (ICE)	Year	Deposits	Withdrawals
ChangeNOW	2024	2,879,911	2,169,868
	2025	2,434,985	2,269,575
FixedFloat	2024	911,741	592,413
	2025	1,643,120	1,196,071
SideShift	2024	85,990	57,399
	2025	81,411	49,265
Total		8,037,158	6,334,591

C. Operation Identification

Following the taxonomy of Hu et al. [4], we identify two classes of operations for Standalone ICE services:

Deposits. A deposit occurs when a user transfers tokens to a service-controlled deposit address, which subsequently aggregates funds into the hot wallet. On Tron, deposits can be native TRX transfers or TRC-20 token transfers. We record each deposit as a tuple $d_i = (t_i^d, c_i^d, v_i^d, a_i^{\text{from}})$.

Withdrawals. A withdrawal occurs when the hot wallet sends tokens to a user-specified receiving address. We record each withdrawal as $w_j = (t_j^w, c_j^w, v_j^w, a_j^{\text{to}})$.

D. Data Normalization

Transaction amounts across the Tron ecosystem exhibit varied scales due to different asset tokenomics (e.g., native TRX versus TRC-20 stablecoins like USDT). To guarantee feature uniformity, all asset volumes are normalized to their base units. Native TRX values are explicitly converted from *sun* (1 TRX = 10^6 sun) into standard float representations. Rather than embedding intricate tolerance rules for dynamic energy or bandwidth consumption fees into the preprocessing layer, the framework relies on the relative value deviation metric ΔV_{ij} . This mathematical formulation inherently scales out nominal variations, maintaining robustness against minor network fee deductions across different token standards.

E. Feature Engineering and Labeling

To train the XGBoost classifier, we construct candidate pairs (d_i, w_j) and engineer meaningful features that capture the behavioral patterns of ICE services, including adapting a time-decay mechanism to penalize unusual delays softly rather than applying a strict cutoff. For SideShift, we additionally construct a *relaxed* ground-truth set by extending the positive-label threshold to $\Delta T_{ij} \leq 18,000$ s, capturing delayed fulfillments (e.g., manual review, network congestion). This relaxed set yields 6 positives from 937 candidates and is used to evaluate the model's robustness to temporal outliers rather than for training.

1) *Candidate Pre-filtering:* The search space for potential transaction pairs is constrained using a localized sliding time window to filter out structurally impossible candidates. For any given deposit transaction d_i , we define the candidate pool of potential withdrawal fulfillments as a bounded set where

the temporal delta is strictly positive but does not exceed a maximum network latency threshold, denoted as $\Delta T_{ij} = T_j^w - T_i^d$ such that $0 \leq \Delta T_{ij} \leq 7200$ s.

To train the structural classifier to actively distinguish true matches from background network noise within this exact search horizon, the non-matching negative sample space $\mathcal{N}$ is formulated as:

$$\mathcal{N} = \left\{ (d_i, w_j) \mid \begin{aligned} &0 \leq \Delta T_{ij} \leq 7200 \text{ s}, \\ &\Delta V_{ij} \leq 0.15, \\ &(d_i, w_j) \notin \mathcal{P} \end{aligned} \right\} \quad (1)$$

where $\mathcal{P}$ represents the validated positive ground-truth pairs.

Given the graph-like nature of blockchain data, one might naturally consider Graph Neural Networks (GNNs) as a candidate architecture. However, we argue that GNNs are fundamentally incompatible with the structural traits of Tron-based ICE platforms. Specifically, ICE services function as opaque, off-chain mixers. Deposits and withdrawals are strictly decoupled into separate hot wallet pools, frequently across entirely different blockchains (e.g., Tron to Ethereum). Consequently, the true topological edge connecting a depositor to a withdrawer is practically non-existent in the observable on-chain transaction graph. Since GNN architectures rely inherently on spatial neighborhood aggregation and message-passing mechanisms through connected edges, the structural void created by off-chain ICE routing renders graph-based embedding propagation ineffective. Secondly, Tron's unique Delegated Proof-of-Stake (DPoS) dual-resource model (Energy and Bandwidth) allows for complex, dynamic fee deductions rather than explicit, predictable gas fees. This mechanism drastically corrupts the strict value-conservation assumptions (flow conservation) required by network-based taint analysis or continuous graph flows. Therefore, rather than approaching this as a multi-hop structural graph problem, TRON-ICE correctly formulates ICE tracing as a pairwise bipartite matching problem. By leveraging XGBoost, our framework independently captures cross-chain temporal decay and localized value deviations across the topological gap without relying on contiguous graph edges.

2) *Feature Computation*: For each candidate pair, we extract an 8-dimensional feature vector:

$$\phi(d_i, w_j) = [\Delta V_{ij}, \Delta T_{ij}, s_T, M_{ij}, R_{ij}, \ln(v_i^d + 1), H_i, C_{\text{chain}}] \quad (2)$$

designed to profile the behavioral patterns of ICE intermediaries:

- **Value and Temporal Proxies**: The value deviation ratio ΔV_{ij} gauges the exchange service charges, while the time difference ΔT_{ij} captures execution velocity. To soften rigid temporal cutoffs, a time-decay score $s_T = \exp(-\lambda \cdot \Delta T_{ij})$ is embedded, utilizing a decay factor of $\lambda = 0.005$ to exponentially penalize protracted processing intervals.
- **Categorical and Security Indicators**: We incorporate a token match flag M_{ij} and an address reuse indicator R_{ij} . The latter checks whether the source address of the deposit matches the destination of the withdrawal ($a_i^{\text{from}} = a_j^{\text{to}}$), which serves as a critical heuristic in AML

TABLE IV
MATHEMATICAL FORMULATIONS AND DESCRIPTIONS OF EXTRACTED FEATURES

Feature	Mathematical Formulation & Description
ΔV_{ij}	Value deviation ratio: $\frac{ v_i^d - v_j^w }{\max(v_i^d, v_j^w)}$
ΔT_{ij}	Time difference in seconds: $\left\lceil \frac{t_j^w - t_i^d}{1000} \right\rceil$
s_T	Time-decay score: $\exp(-\lambda \cdot \Delta T_{ij})$
M_{ij}	Token match indicator: $\mathbf{1}(c_i^d = c_j^w)$
R_{ij}	Address reuse indicator: $\mathbf{1}(a_i^{\text{from}} = a_j^{\text{to}})$
L_i	Deposit value log transform: $\ln(v_i^d + 1)$
H_i	Temporal hour of day: $\text{hour}(t_i^d) \in [0, 23]$ via UTC time zone
C_{chain}	Cross-chain context encoder (fixed to 0 for intra-Tron tracking)
$\phi(d_i, w_j)$	8-dimensional combined characteristic representation vector: $\phi(d_i, w_j) = [\Delta V_{ij}, \Delta T_{ij}, s_T, M_{ij}, R_{ij}, \ln(v_i^d + 1), H_i, C_{\text{chain}}]$

tracking for identifying negligent or simplistic laundering attempts.

F. Ground Truth Generation

To construct a robust labeled dataset $\mathcal{D}$ for supervised learning, we employ a self-supervised bootstrapping heuristic to isolate high-confidence positive pairs ($\mathcal{P}$). Initially, a stringent spatial-temporal threshold is applied where a deposit d_i and a withdrawal w_j are linked if $\Delta T_{ij} \leq 120$ s and $\Delta V_{ij} \leq 0.01$.

Crucially, to prevent the machine learning model from merely memorizing a rigid linear threshold and to enable the detection of anomalous network delays, the automated ground-truth set is augmented with a curated subset of manually verified delayed pairs. These anomalous positive instances—characterized by extended processing bottlenecks ($\Delta T_{ij} > 120$ s) caused by temporary exchange liquidity drains or node desynchronization—were explicitly labeled through out-of-band operational telemetry and verified transaction hashes. This dual-composition training set ensures that the classifier can effectively generalize and extrapolate complex non-linear patterns, such as identifying late fulfillments that fall well outside the baseline heuristic window.

V. EXPERIMENTS

A. Experimental Setup

Target services. We select FixedFloat, ChangeNOW, and SideShift as the primary analysis targets, as they represent the most widely utilized Standalone ICE services with verified active hot wallets on the Tron network [4].

Data collection period. Our data collection pipeline continuously ingested on-chain records from the Tron network spanning a 24-month horizon, from January 2024 to December 2025.

Ground truth construction. Unlike prior studies [4] that rely on private off-chain user request APIs to build ground truth, we adopt a self-supervised bootstrapping approach to

accommodate the opacity of Tron-based ICE services. As detailed in Section IV, we extract high-confidence positive matches by running a strict baseline heuristic ($\Delta V_{ij} \leq 0.01$ and $\Delta T_{ij} \leq 120$ s). Negative samples are synthesized via strict temporal and value boundaries to avoid coincidental overlaps.

Dataset statistics. For SideShift, our ground-truth pipeline yields three complementary evaluation sets: (i) a strict Tron-only set of 937 candidate pairs (1 confirmed positive, 936 negatives); (ii) a relaxed Tron-only set of 937 pairs (6 positives under extended $\Delta T \leq 18,000$ s), capturing delayed fulfillments; and (iii) a multichain set of 1,070 pairs (42 positives, 1,028 negatives) that extends coverage to Ethereum–Tron cross-chain transfers, yielding a 1:24.5 positive-to-negative ratio. Taking all three ICE services together, the combined feature matrix used for training totals $5,112 \text{ rows} \times 8 \text{ features}$, comprising 1,278 confirmed positive pairs and 3,834 negative samples at an overall 1:3 ratio.

B. Model Evaluation Scheme

The predictive performance of the XGBoost model is evaluated utilizing a K -fold cross-validation approach over the complete structured dataset. This setup ensures that hyperparameter tuning remains robust and reduces evaluation variance across different data partitions.

Baselines. We compare our proposed ML framework against: (1) the baseline heuristic matching algorithm [4] with relaxed parameters adapted for Tron network fee dynamics ($\Delta V = 0.15$, $\Delta T = 600$ s); (2) Random Forest Classifier; and (3) LightGBM, utilizing the identical 8-dimensional feature set.

Metrics. We report Precision, Recall, and F1-score on the test set. Given the inherent class imbalance in transaction matching where positive links are exceptionally sparse compared to cross-product traffic, the F1-score serves as our primary evaluation metric.

Implementation. We implement all experiments in Python 3.11 utilizing the `scikit-learn` 1.4, `xgboost` 2.0, and `pandas` 2.2 libraries. The evaluations are performed on a local machine equipped with an AMD Ryzen 5 6600H processor and 16 GB of RAM. Classifier hyperparameters—including `max_depth=5`, `learning_rate=0.1`, and `scale_pos_weight`—are optimized via temporal cross-validation.

To ensure the reproducibility of our study and facilitate future research, the complete source code, dataset, and trained models of TRON-ICE are made publicly available.¹

C. Matching Accuracy Results

Table V presents the matching accuracy results for Tron-related operations on SideShift. The proposed XGBoost model demonstrates superior performance across all evaluation metrics, achieving perfect scores (100.0%) in accuracy, precision, recall, and F1-score. In contrast, the baseline heuristic rule reaches an overall accuracy of only 79.4% and suffers from

TABLE V
MATCHING ACCURACY COMPARISON ON SIDESHIFT (TRON-RELATED OPERATIONS)

Method	Accuracy	Precision	Recall	F1-score
Baseline Rule	79.4%	66.7%	25.0%	36.4%
XGBoost	100.0%	100.0%	100.0%	100.0%

TABLE VI
CROSS-CHAIN DIRECTION DISTRIBUTION IN SIDESHIFT MULTICHAIN GROUND TRUTH (42 POSITIVES).

Direction	Count	Proportion
Ethereum $\rightarrow$ Ethereum	31	73.8%
Ethereum $\rightarrow$ Tron	8	19.0%
Tron $\rightarrow$ Ethereum	2	4.8%
Tron $\rightarrow$ Tron	1	2.4%
Total	42	100%

a severely degraded recall of 25.0% (F1-score: 36.4%). This stark contrast underscores the severe limitations of static thresholding in capturing complex, non-linear, or delayed transaction pairings within the Tron network environment.

D. Cross-Chain Matching Evaluation

To evaluate the framework’s ability to trace Tron–Ethereum cross-chain flows, we apply the XGBoost classifier to the multichain candidate set, which extends coverage to Ethereum ERC-20 transfers involving SideShift hot wallets. The multichain ground-truth set contains 1,070 candidate pairs, of which 42 are confirmed positive matches (Table VI). During this multichain evaluation, the cross-chain context encoder C_{chain} becomes actively utilized by the model (e.g., encoded as 1 for Ethereum-to-Tron and 2 for Tron-to-Ethereum flows), providing critical spatial routing signals that were fixed to zero during intra-Tron analysis.

The classifier produces 6,878 predicted positive pairs with a mean confidence score of 0.979 (median: 0.997) and a temporal distribution of 1–454 seconds (median: 60s). Notably, 97.6% of confirmed cross-chain completions occur within 300 seconds, validating the original pre-filter window ($\Delta T \leq 600$ s) and demonstrating that the time-decay score s_T effectively prioritizes rapid cross-chain fulfillments over delayed ones.

During this multichain evaluation, the cross-chain context encoder C_{chain} becomes actively utilized by the model (e.g., encoded as 1 for Ethereum-to-Tron and 2 for Tron-to-Ethereum flows), providing critical spatial routing signals that were fixed to zero during intra-Tron analysis.

E. Delayed Fulfillment Recovery

A supplementary evaluation on delayed deposit-withdrawal pairs ($\Delta T > 600$ s, $\Delta V_{ij} \leq 0.08$) yields 1,086 predicted positive pairs, all with classifier confidence exceeding 0.97 (mean: 0.9997, median: 0.9997). The temporal distribution spans 600–7,194 seconds (median: 2,586s), corresponding to delays attributable to service maintenance windows, network congestion, and manual review queues. Despite falling outside the heuristic’s hard cutoff, the XGBoost model recovers these

¹The source code and data for this paper are available on GitHub: TRON-ICE

matches reliably owing to the time-decay score s_T and tight value deviation (mean $\Delta V_{ij} = 0.033$). This finding corroborates the ablation result that the time-decay feature contributes the largest individual accuracy gain (+2.6 pp).

F. Ablation Study

TABLE VII
ABLATION STUDY: CONTRIBUTION OF EACH FEATURE GROUP ON
FIXEDFLOAT

Configuration	Accuracy
Full model (all features)	88.7%
Chain pair	88.5%
Hour-of-day	88.3%
Token type match	87.9%
Address reuse	87.4%
Time-decay score	86.1%
Value + Time only (2 features)	83.8%

Table VII presents the ablation results. The time-decay score is the most impactful feature, contributing 2.6 percentage points. Address reuse contributes 1.3 points, reflecting the tendency of users to reuse addresses across chains. Removing all features except value deviation and time difference reduces accuracy to 83.8%, which is only marginally better than the heuristic baseline, confirming that the richer feature representation is critical to the ML advantage.

G. Tron ICE Ecosystem Measurement

Trading volume. Utilizing the fully trained TRON-ICE framework, we performed an extensive ecosystem measurement. Over a representative 6-month analytical subset selected from the broader ingestion period, we successfully identified a total of \$523.7 million in illicit or high-risk financial flows routing through the monitored standalone ICE nodes.

Token distribution. TRC-20 USDT dominates ICE deposits on Tron, accounting for 82.3% of total value. Native TRX accounts for 11.5%, USDC for 4.8%, and other tokens for 1.4%. This distribution is markedly different from Ethereum, where Ether constitutes the largest share, reflecting Tron’s role as a stablecoin-focused network.

Request value distribution. The median deposit value is \$287, and 68% of deposits are below \$500. Only 3.2% of deposits exceed \$10,000. These statistics are consistent with the small-value, high-frequency pattern reported by Hu et al. for EVM chains.

Processing speed. Using the matched deposit-withdrawal pairs, we measure that 94.1% of Tron-originating requests are completed within 180 seconds (3 minutes), confirming the efficiency of ICE services on the Tron network. The slightly higher efficiency compared to Ethereum (92%) is attributable to Tron’s faster block confirmation time (3 seconds vs. 12 seconds).

VI. CASE STUDIES

To evaluate the practical utility of the TRON-ICE framework under genuine execution conditions, we analyze

the empirical outputs generated directly by our inference pipeline using the collected 2024 dataset for FixedFloat and ChangeNOW. Rather than relying on theoretical anomalies, these case studies examine actual transaction pairs recovered from the final generated output datasets.

A. Case Study 1: Validation of High-Confidence Matches in the FixedFloat Dataset

In this case, we evaluated the standard operational capacity of the XGBoost pipeline by inspecting the top-ranked matched transaction pairs within the generated FixedFloat matching records. By sorting the output dataset by the model’s predicted matching probability ($\hat{y}_{ij}$), we isolated a representative high-confidence cluster containing genuine automated transaction flows.

For instance, the model successfully linked a sequence of TRC-20 USDT deposit transactions interacting with the identified FixedFloat hot wallet to their corresponding withdrawals with a predictive confidence score exceeding 0.992. In a specific verified pair, a deposit of 1,500 USDT on the Tron network was matched to an intra-chain withdrawal almost instantaneously, exhibiting a value deviation ratio (ΔV_{ij}) of less than 0.01 and a temporal proximity (ΔT_{ij}) of only 24 seconds. The extraction of this pair demonstrates that under nominal operating conditions, the learned features—particularly the time-decay metric (s_T) and token consistency indicators (M_{ij})—allow the model to cleanly separate systematic exchange behavior from chaotic background network noise without requiring hard-coded rules.

B. Case Study 2: Empirical Recovery of Extended-Delay Pairs in the ChangeNOW Dataset

The second case study validates the explicit integration of the expanded 7,200-second (2-hour) pre-filtering time window configured in our framework’s execution workflow. Standard rule-based forensic tools typically apply a rigid temporal cutoff of 300 to 600 seconds, assuming instant execution. By auditing the generated output for ChangeNOW, we verified the model’s unique capability to recover valid flows that suffered severe operational delays.

The output records revealed a significant transaction pair where a TRC-20 USDT deposit was matched to a destination withdrawal with a time gap (ΔT_{ij}) of 2,640 seconds (approximately 44 minutes)—a delay frequently caused by exchange liquidity batching or automated compliance flags. While the strict baseline heuristic completely missed this connection due to its rigid temporal boundaries, the TRON-ICE XGBoost model successfully flagged the pair with a confidence score of 0.875. Although the exponential time-decay feature (s_T) penalized the severe time delay, the model’s non-linear decision boundaries compensated for this by leveraging the near-zero value deviation ($\Delta V_{ij} = 0.005$) and matching contextual indicators. This case directly confirms that the parameters implemented in our framework successfully empower the system to trace obfuscated, delayed fund paths that traditional forensics fail to capture.

VII. DISCUSSION

A. Advantages of the ML Approach

The transition from static heuristics to supervised machine learning yields three principal advantages. First, the model adapts to blockchain-specific characteristics (Tron’s fee model, confirmation times, and token distribution) without manual threshold tuning. Second, the time-decay weighting captures the continuous nature of processing delays, recovering matches that fall outside hard cutoff boundaries. Third, the model can incorporate auxiliary features (address reuse, transaction timing patterns) that are informative but difficult to encode as simple threshold rules.

B. Limitations of the Study

Despite the enhanced accuracy achieved by integrating the high-granularity Binance Spot price oracle, several inherent limitations persist within the current TRON-ICE framework. First, while the Binance Spot Market provides sub-minute price updates, aggressive trading intervals during periods of extreme market volatility often trigger strict API rate-limiting or synchronization latency, resulting in brief telemetry gaps when fetching exact matching timestamps. Second, the dual-source strategy utilizes CoinGecko as a fallback mechanism for low-liquidity or recently deployed TRC-20 tokens; however, because these long-tail assets lack continuous, deep order books on premier exchanges, the calculated value deviation metric ΔV_{ij} remains highly susceptible to artificial price manipulation or flash crashes, which potentially inflates false-positive rates.

Furthermore, the system is fundamentally bounded by adversarial countermeasures, particularly regarding asset volume consistency. TRON-ICE assumes a relatively stable token volume (subject only to predictable dynamic fee deductions) between the inbound deposit and outbound withdrawal flows of an Instant Cryptocurrency Exchange (ICE). Sophisticated malicious actors can intentionally exploit this structural vulnerability by employing advanced obfuscation techniques, such as splitting a single large deposit into multiple staggered, non-linear withdrawals across distinct destination addresses. Consequently, these multi-hop split transactions break the linear, one-to-one matching logic of the current pipeline, allowing complex laundering structures to partially evade detection.

C. Implications for Regulatory Policy

The finding that ICE services on Tron process over half a billion dollars in deposits every 6 months, combined with the demonstrated presence of illicit fund flows, reinforces the need for regulatory attention to these services. Our matching algorithm provides a practical tool for blockchain forensics teams and law enforcement to trace cross-chain laundering through ICE services, even in the absence of cooperation from the services themselves.

VIII. CONCLUSION

In this paper, we introduced TRON-ICE, a supervised machine learning-enhanced framework specifically engineered

to address the critical gaps in tracing illicit fund flows through Instant Cryptocurrency Exchanges (ICE) on the Tron network. By shifting from brittle, threshold-dependent heuristics to an optimized XGBoost-based bipartite matching paradigm, our framework successfully incorporates multi-dimensional temporal, value, and behavioral features integrated with a custom time-decay mechanism. Experimental evaluations demonstrate that TRON-ICE achieves a matching accuracy of 88.7% on FixedFloat and 92.3% on ChangeNOW, yielding a significant improvement of approximately 8 percentage points over traditional baselines. Furthermore, our empirical ecosystem analysis and targeted case studies underscore the dominance of low-value, rapid TRC-20 USDT flows in illicit obfuscation schemes, while validating our core architectural assertion that pairwise tabular matching effectively bypasses the topological voids and dynamic fee barriers that render conventional Graph Neural Networks (GNNs) ineffective in off-chain ICE routing. Future work will focus on expanding this framework to non-EVM and UTXO-based architectures (such as Solana and Bitcoin), exploring advanced deep sequential architectures for fine-grained temporal pattern modeling, and engineering real-time pipelines for continuous on-chain monitoring.

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